Computers and Electronics in Agriculture 173 (2020) 105401

Contents lists available at ScienceDirect

Computers and Electronics in Agriculture

journal homepage: www.elsevier.com/locate/compag

Field path planning using capacitated arc routing problem

Amin Khajepour[a], Majid Sheikhmohammady[a,b,*], Ehsan Nikbakhsh[a]

[a] Faculty of Industrial and Systems Engineering, Tarbiat Modares University, Tehran, Iran
[b] Conflict Analysis Group, Department of Systems Design Engineering, University of Waterloo, Canada

ARTICLE INFO

Keywords:
Adaptive large neighborhood search
Agricultural machinery
Capacitated arc routing problem
Field path planning

ABSTRACT

Duration of Agricultural operations is a crucial issue in fleet management as it directly affects the operation costs. Agricultural fleet usually traverses paths, covering the whole field to complete the operation. The agricultural operations are not necessarily performed efficiently due to the various variables including field shape, presence of obstacles in the field, number of vehicles, and their specifications. In this study, agricultural operations are modeled as a capacitated arc routing problem (CARP). CARP is a combinatorial optimization problem, which determines the harvest paths in a field and finds the best order of traversing. By dividing the paths into two categories of required and non-required, CARP demonstrates a field as a graph. Using CARP, no extra or invalid edge on the graph would be needed. As CARP is an NP-Hard problem, a solution algorithm based on adaptive large neighborhood search (ALNS) is employed to solve large instances of the problem under study. The proposed ALNS algorithm is applied to well-known problem instances as well as a realistic case study to achieve the optimal harvesting pattern efficiently. The computational results demonstrate that the ALNS algorithm surpasses the conventional harvesting pattern. The amount of improvement obtained by the proposed method could vary due to the several factors, including the depot location, vehicle capacity, and field shape.

1. Introduction

Most of the agricultural operations, such as harvesting, spraying, and seeding, are performed using agricultural machinery. For large scale operations, it is often necessary to hire a fleet of machines to complete the operations on time. The required time to fulfill the agricultural operations depends on how machines are managed. The completion time is important for the farmers as it directly affects the operation costs and therefore, the final price of the products. Thus, fleet management has a significant role in preventing extra expenses during operations.

Machine working time consists of operation time and non-productive time. Operation time is defined as the time required by a machine to perform its special duty on the field like harvesting, mowing, trilling, and spraying. The non-productive time is the time that a machine spends moving across the headland, maneuvering, and loading or unloading the hopper. To improve the working time efficiency of the machines, it is recommended to minimize the non-productive time (Srivastava et al., 1993). To achieve this goal, different operations research methods have been proposed by researchers such as coverage path planning problems (Oksanen and Visala, 2009), routing problems (Bochtis and Sørensen, 2009), as well as simulation (Zhou et al., 2015).

Farmers may not use operations research methods due to several reasons. First, in many countries, traditional farmers manage most fields using old-fashioned methods based on the farmers' judgment and experience (Jian et al., 2016). Second, agricultural operations such as harvesting and spraying have simple, intuitive solutions that could perform well enough on small-sized fields that have a small economy of scale. Such solutions do not necessarily perform well on large-sized fields. Nevertheless, the farmers may not use operations research methods to find the best possible solutions because they underestimate the improvements that could be achieved. They might believe that the cost saving obtained via optimization could not be sufficient to cover analysis and consultancy costs. However, operations research could take into account the intricacies of the problem and yield better results (i.e., less vehicle costs as well as fuel consumption). In addition, as autonomous vehicles become more common in agriculture, operations research techniques would be more applicable in planning the farm operations. Finally, agricultural economists could use operations research as a useful tool for analyzing various agricultural decisions and policies (Arfini et al., 2016, p. 14).

In this study, the arc routing problem (ARP) approach is used to optimize the paths that agricultural machines traverse in the field. Routing problems belong to a class of combinatorial optimization

* Corresponding author at: Faculty of Industrial and Systems Engineering, Tarbiat Modares University, Tehran, Iran.
E-mail address: msheikhm@modares.ac.ir (M. Sheikhmohammady).

https://doi.org/10.1016/j.compag.2020.105401
Received 28 August 2019; Received in revised form 23 February 2020; Accepted 31 March 2020
Available online 15 April 2020
0168-1699/ © 2020 Elsevier B.V. All rights reserved.

problems that have various applications in logistics. Routing problems seek to satisfy all the customers' demands at the minimum cost. In routing problems, there are some customers with positive demands that are served by a fleet of vehicles. Responding to customers' demands requires traversing multiple paths, which in turn imposes traveling costs to the system. Each vehicle is stationed at a predetermined depot (or depots) and ready to meet the customers' demands through a road network, which could be depicted as a graph. Each vehicle makes a tour by beginning from the depot, serving a subset of customers while observing its capacity, and then returning to the depot. The main objective of routing problems is to determine the best tours with the minimum traveling cost. The routing problems include two major classes of problems, which are different in how to deal with the demands on the graph.

In the first class, entitled vehicle routing problem (VRP), customers' demands are considered on vertices of a graph. In VRP, customers and depots are located on the vertices of the graph. The edges show the path between the vertices and their associated costs. This problem has real-world applications such as goods and drink distribution, waste collection, as well as school bus routing. Some other extensions of the problem are vehicle routing problems with time windows, split-delivery VRP, and VRP with backhauls.

The second class of routing problems is called the arc routing problem (ARP). In this problem, the customers and their demands are considered on the edges of the graph. Each edge has a non-negative amount of demand which must be served by vehicles. The edges that have positive demands are called *required* edges or *tasks*. The remaining edges are called non-required or no-service edges (Corberán and Laporte, 2013). In addition, the cost of each edge is predetermined. The goal of arc routing problem is to determine a set of tours for the vehicles so that traversing all the required edges be done at the minimum cost. *Meter reading, snow plowing, salt spreading*, and *road maintenance* are some of the applications of this problem (Corberán and Laporte, 2013). Other extensions of CARP have been considered in the literature including but not limited to multi-depot CARP, multi-compartment CARP, and open CARP.

Agricultural operations, such as harvesting, planting, and spraying, are intrinsically logistics problems. In these operations, it is necessary to transport a bulk of material from the warehouse to the farm or vice versa. Thus, routing problems can be applicable in modeling these operations. These models could increase the efficiency of the agricultural fleets. While both classes of routing problems could be applied to modeling agricultural operations, these operations are inherently similar to arc routing problems. The advantage of ARP is that there is no need for major changes in data preparation and model building.

Contrary to the current literature on the application of VRP models to agricultural operations, this research focuses on modeling the agricultural operations carried out by machines using an ARP approach to reduce the total cost/time of such operations. Due to transferring agricultural products in such operations, it is necessary to take into account the capacity of the machines. Therefore, the capacitated arc routing problem (CARP) is an appropriate approach to model agricultural operations. CARP can be used in many agricultural operations like harvesting, seeding, and spraying. CARP was introduced by Golden and Wong (1981) and proved to be an NP-hard problem (Corberán and Laporte, 2013). Hence, obtaining optimum results in polynomial time is impossible. Metaheuristic algorithms could be employed to obtain high-quality solutions in a reasonable amount of time. Inspired by the notable performance of the adaptive large neighborhood search in routing problems, this method is employed to solve the agricultural path planning problem.

The remainder of the article is organized as follows. Section 2 presents the literature review and previous methods. The problem definition is given in Section 3. In Section 4, the CARP mathematical model is presented. In Section 5, the adaptive large neighborhood search is described. The computational results on CARP instance problems and a realistic example filed are reported in Section 6. Finally, sensitivity analysis and conclusion are presented in Sections 7 and 8 respectively.

2. Literature review

In order to optimize the infield and inter-field agricultural operations, researches have used different approaches. These approaches could be categorized into three categories, namely coverage path planning, linear programming, and routing problems. It is worth mentioning that combinations of these approaches have also been used in some studies.

The path planning problem is a practical approach for covering the whole field surface. It has many applications in robotics as well. This method was suggested by Oksanen and Visala (2009) to optimize the traversing routes of agricultural machines in the field. Their method aims to find the best route in complex-shaped fields by splitting them into simple-shaped ones using greedy algorithms. Spekken and de Bruin (2013) proposed a path planning method that minimizes non-productive times, including maneuvering and servicing times of agricultural machinery. The orientation of the tracks (crop rows), the order of tracks, and two types of turns between tracks were considered. To find the best order of traversing all the tracks, the traveling salesman problem (TSP) is employed that was solved using the Clarke-Wright savings algorithm. Also, Zhou et al. (2014) used the path planning problem and traveling salesman problem to tackle the presence of field obstacles.

Similar to path planning, the VRP can cover the whole field surface and determine the optimum order of field tracks. VRP is the extended version of the multiple TSP that considers vehicle's capacity constraint too. Bochtis and Sørensen (2009) discussed how VRP and its other extensions could be implemented for various agricultural operations. To perform harvesting operations, a combination of vehicle routing problem and minimum-cost flow network problem was proposed by Ali et al. (2009). To achieve this, the field is gritted and each pixel is assumed as a customer of VRP model, then in order to prevent prohibited turns some penalty constraints were added to the model. This model is solved by the ILOG CPLEX, which could not tackle large instances efficiently.

Based on the application, multiple objectives are considered in some other studies. Herrera-Cáceres et al. (2017) considered a multi-objective mixed-integer linear model to optimize the harvesting operation to maximize olive oil extraction. Sethanan and Neungmatcha (2016) introduced a multi-objective model that considered maximizing sugarcane yield and minimizing total distance. Particle Swarm Optimization was employed to solve the model.

Later studies implemented different metaheuristics to obtain better results in a shorter time. Seyyedhasani and Dvorak (2017) implemented CVRP to reduce the operations time considering multiple vehicles. The Modified Clarke-Wright (MCW) and tabu search (TS) were proposed as the solution methods. The computational results demonstrated that the proposed tabu search provided better results than MCW. Later, these methods were implemented on various real fields with different scenarios in order to compare the field efficiency of the presented routes (Seyyedhasani et al., 2019). Pitakaso and Sethanan (2019) proposed a mathematical model to optimize sugarcane harvesting, considering the prespecified number of harvesters and time windows. The model's objective is to maximize the sugarcane field coverage by harvesters. It is solved using the ALNS algorithm.

Contrary to the previous studies, Santoro et al. (2017) implemented a different type of routing problem, namely the rural postman problem (RPP), for sugarcane harvesting operation to minimize the agricultural operation time. RPP is the basic form of the arc routing problem. RPP consists of several required and non-requires edges. However, contrary to the CARP, there is no capacity constraint in RPP. Thus, it is applicable in cases that the edges are served by uncapacitated vehicles.

So far, different approaches suggested for agricultural fleet planning in order to complete the operations more efficiently are presented. Due to the logistical nature of agricultural operations, the capacitated arc routing problem, as an optimization tool, is introduced in this study to perform the agricultural operations. Unlike the vehicle routing problem, which assumes that the field could be represented as a complete graph, the arc routing problem plots the field graph based on the shape of the field and the crop rows. Therefore, there would be no invalid or extra edges which needed to be dealt with while using ARP approaches. To the best of our knowledge, capacitated arc routing problem (CARP) has never been used for optimizing the agricultural operations.

3. Problem definition

The agricultural operations, such as harvesting, spraying, and seeding, are accomplished using one or several tractors, harvesters, or robots. Completing an operation usually involves transferring material (i.e., products, chemicals, or seeds) from warehouses or silos to the field and vice versa. In this study, several assumptions are considered to perform an operation as follows:

- The crop rows are predetermined considering the field characteristics (i.e., field slope, geographical position, etc.)
- The agricultural machinery are assumed homogenous having an identical capacity and working width.
- Only one depot is used to accomplish the operation.

Thus, an agricultural operation is a logistics problem that could be conducted more efficiently using routing problems. As these operations are inherently similar to the arc routing problem, applying ARP to them is straightforward. For demonstrating an operation as a capacitated arc routing problem. It is just needed that the agricultural operation's components like agricultural machinery, crop rows, field's yield, field's paths, and warehouse's locations, are mapped to the arc routing problem's components (i.e., vehicles, the set of vertices and edges, the cost and the demand of each edge, and the location of the depots, respectively). First, the graph of routing problem must be obtained according to the field's map. The tracks that agricultural machinery traverse in the field are assumed as the edges of the graph and the endpoints of these tracks are shown by the vertices. In addition, the location of the warehouses or silos or other facilities, used in an agricultural operation, are indicated by vertices called depot vertices.

The tracks that the agricultural machinery traverse to fulfil their special duties (e.g., harvesting, spraying or seeding) are assumed as the required edges that must be traversed at least once. Thus each required edge is determined according to the implement's effective working width. The other tracks usually traversed for maneuvering, like tracks at headlands, are assumed as no-service or unrequired edges. Also, the repetitive traversals of required edges are assumed as no-service edges. The cost associated with each edge represents the distances traveled by the machinery or the time taken for the edge traversal. The demand associated with each edge represents the amounts of harvested crops or yields of the corresponding required edge or the amounts of chemicals needed to be applied on the corresponding required edge.

The agricultural fleets involved in an operation are considered as the vehicles of the arc routing model. The fleet is assumed homogeneous meaning that all vehicles have the same characteristics such as capacity, speed, and working width. Fig. 1 depicts how to implement the ARP model on the in-field agricultural operation.

At first, each vehicle is stationed at the depot, and it makes a tour by starting from the depot, traversing several required edges until the capacity is filled, and then returning to the depot. Loading/unloading the vehicles' hopper or fueling them if needed, is done in the depot. The next step is determining the edges' costs and demands. The costs associated with required or non-required edges are determined based on the traversal time or length of each edge. The amount of demand for each required edge is equal to the amounts of material that must be distributed (e.g., in spraying or seeding operation) in the filed or collected from the field (e.g., harvesting operation). As there is no need to distribute or collect any material from the non-required edges, their demands are considered zero.

4. Mathematical formulation

The general form of CARP is described on an undirected graph. Consider graph $G = (V, E)$, where V is the set of vertices and E is the set of edges. E_r is a subset of E that contains the required edges with positive demand of q_{ij}. Corresponding to each edge a cost c_{ij} is defined to show traversal cost from node i to node j. In asymmetric applications of CARP (i.e., $c_{ij} \neq c_{ji}$), a directed graph is used, thus two arcs with opposite directions are defined instead of each edge. One of the vertices, usually the first node, is assumed as the *depot*. Each vehicle has a limited capacity of W (assuming $W \geqslant \max_{(i,j)\in E}\{q_{ij}\}$ to allow a single visit satisfying the demand of each edge). All vehicles start from the depot, traverse a number of required edges while observing the capacity constraint, and then return to the depot to complete a feasible tour. The required edges are not allowed to be served by split-delivery which is using multiple vehicles or multiple traversals. Repetitive traversals of required edges are considered as non-required edges. The goal of the capacitated arc routing problem is to serve all the required edges at the minimum cost.

In the presented model, n and k, show the number of vertices (nodes) and the number of vehicles, respectively. Variable x_{ij}^p is a binary variable that is equal to 1 if vehicle p traverses edge (i, j). Binary variable l_{ij}^p is equal to 1 if vehicle p services edge (i, j). Finally, $\lceil z \rceil$ equals the smallest integer that is greater than or equal to z.

CARP can be formulated as an integer programming problem as follows (Golden and Wong, 1981):

$$\min \sum_{i=1}^{n}\sum_{j=1}^{n}\sum_{p=1}^{k} c_{ij}x_{ij}^p \tag{1}$$

Subject to:

$$\sum_{k=1}^{n} x_{ki}^p - \sum_{k=1}^{n} x_{ik}^p = 0 \quad \forall\, i = 1, \ldots, n, \ \forall\, p = 1, \ldots, k \tag{2}$$

$$\sum_{p=1}^{k} (l_{ij}^p + l_{ji}^p) = \left\lceil \frac{q_{ij}}{W} \right\rceil \quad \forall\, (i,j) \in E \tag{3}$$

$$x_{ij}^p \geqslant l_{ij}^p \quad \forall\, (i,j) \in E, \ \forall\, p = 1, \ldots, k, \tag{4}$$

$$\sum_{i=1}^{n}\sum_{j=1}^{n} q_{ij} l_{ij}^p \leqslant W \ \forall\, p = 1, \ldots, k \tag{5}$$

$$\left.\begin{aligned} &\sum_{i\in\tilde{Q}}\sum_{j\in\tilde{Q}} x_{ij}^p + n^2 y_{1\tilde{q}}^p \leq \left|\tilde{Q}\right| - 1 && (6)\\ &\sum_{i\in\tilde{Q}}\sum_{j\in\tilde{Q}} x_{ij}^p - y_{2\tilde{q}}^p \geq 1 && (7)\\ &y_{1\tilde{q}}^p + y_{2\tilde{q}}^p \leq 1 && (8)\\ &y_{1\tilde{q}}^p, y_{2\tilde{q}}^p \in \{0,1\} && (9) \end{aligned}\right\} \quad \begin{aligned} &\forall p = 1,\ldots,k, \\ &\tilde{q} = 1,\ldots,2^{n-1}-1, \\ &\forall \tilde{Q} \in V, \left|\tilde{Q}\right| \geq 2 \end{aligned}$$

$$x_{ij}^p, l_{ij}^p \in \{0, 1\} \ \forall\, i = 1, \ldots, n, \ \forall\, j = 1, \ldots, n, \ \forall\, p = 1, \ldots, k \tag{10}$$

The first equation is the objective function that minimizes the total traversed length by vehicles. Eq. (2) is called the continuity constraint, which forces the vehicle to continue the route from its current position (if a vehicle enters a node, then it must leave it). Eq. (3) makes sure the

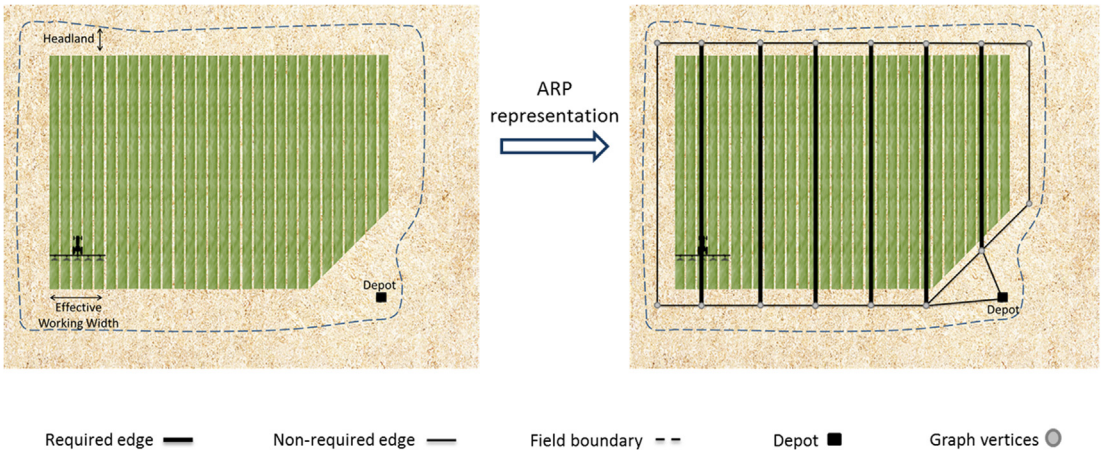


Fig. 1. Implementation of ARP on an Agriculture field.

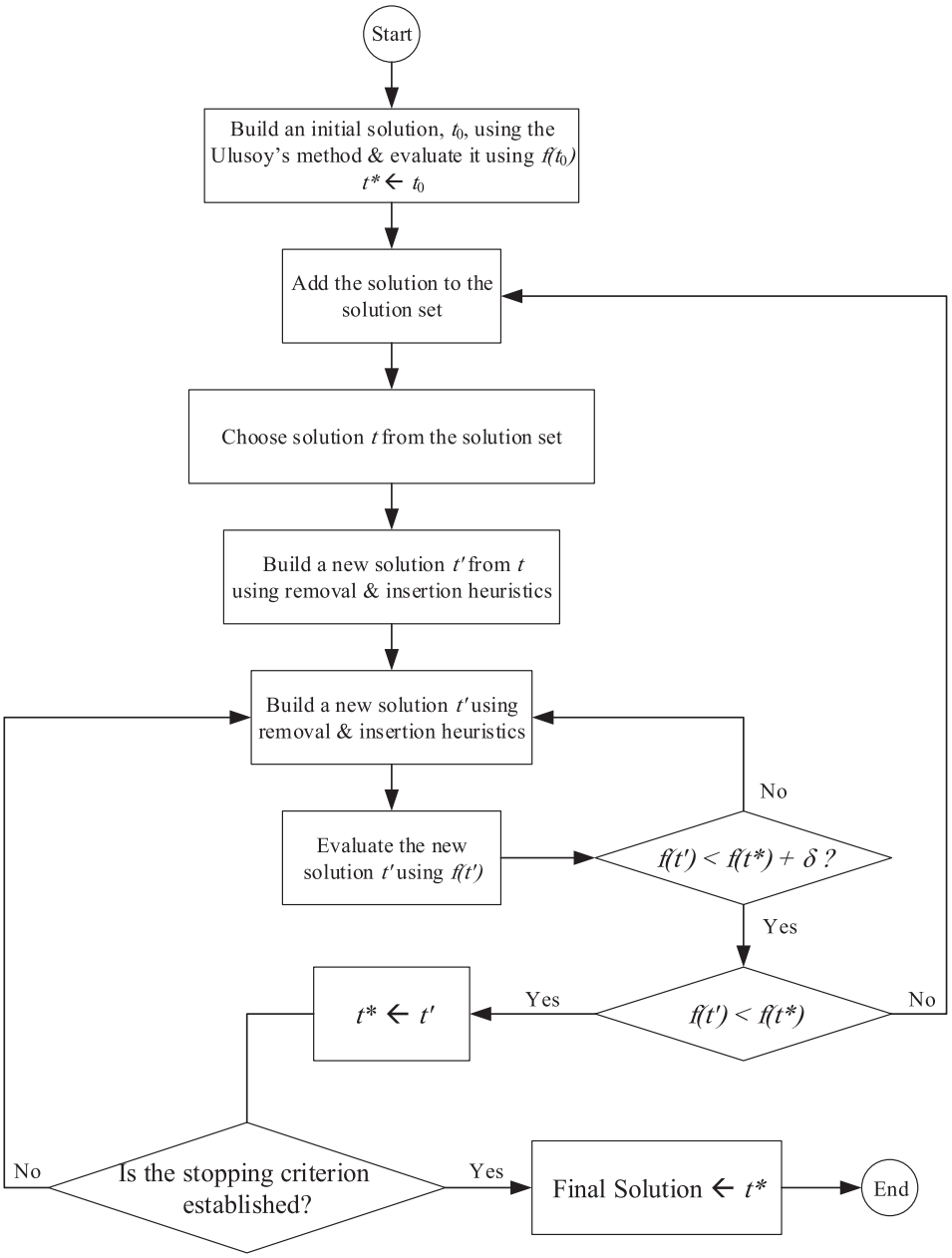


Fig. 2. ALNS algorithm diagram.

Table 1
ALNS results of KSHS instance problems.

Instance no.	Vertices	Required edges	Optimum	ALNS result	% Deviation
KSHS 1	8	15	14,661	14,661	0%
KSHS 2	10	15	9863	9863	0%
KSHS 3	6	15	9320	9320	0%
KSHS 4	8	15	11,498	11,498	0%
KSHS 5	8	15	10,957	10,957	0%
KSHS 6	9	15	10,197	10,197	0%

Table 2
ALNS results of GDB instance problems.

Instance no.	Vertices	Required edges	Optimum	ALNS result	% Deviation
GDB 1	12	22	316	316	0.00
GDB 2	12	26	339	339	0.00
GDB 3	12	22	275	275	0.00
GDB 4	11	19	287	287	0.00
GDB 5	13	26	377	403	6.90
GDB 6	12	22	298	312	4.70
GDB 7	12	22	325	325	0.00
GDB 8	27	46	348	363	4.31
GDB 9	27	51	303	325	7.26
GDB 10	12	25	275	285	3.64
GDB 11	22	45	395	421	6.58
GDB 12	13	23	458	474	3.49
GDB 13	10	28	536	579	8.02
GDB 14	7	21	100	104	4.00
GDB 15	7	21	58	62	6.90
GDB 16	8	28	127	133	4.72
GDB 17	8	28	91	97	6.59
GDB 18	9	36	164	177	7.93
GDB 19	8	11	55	55	0.00
GDB 20	11	22	121	124	2.48
GDB 21	11	33	156	170	8.97
GDB 22	11	44	200	216	8.00
GDB 23	11	55	233	252	8.15

required edges are served just in one traverse. Eq. (4) ensures that the edge (i, j) can be served by vehicle p only if that vehicle traverses this edge. Eq. (5) is the vehicle capacity constraint. Eqs. (6)–(8) are the sub-tour elimination constraints (i.e., each tour must include the depot). Finally, Eqs. (9) and (10) define the decision variables as binary variables.

5. Solution method

As previously mentioned, CARP is an *NP*-hard problem. Hence, obtaining its optimal solution for large instances using exact algorithms (such as branch & bound) could take a long time and even sometimes is impossible. In this study, the adaptive large neighborhood search (ALNS) algorithm is applied to the problem, to obtain high quality solutions. Proposed by Ropke and Pisinger (2006), ALNS is an improvement algorithm that starts with an initial solution and improves it in each iteration to find the optimal solution. ALNS uses several removal and insertion heuristics. In addition, ALNS selects heuristics in each iteration via determining each heuristic weight in an adaptive way (Ropke and Pisinger, 2006).

5.1. Initial solution

To find the optimal solution, ALNS needs an initial solution as its input. There are some heuristics introduced to solve other variants of ARP such as Rural Postman Problem (RPP), Chinese Postman Problem (CPP) and Capacitated Chinese Postman Problem (CCPP). According to CARP literature, these heuristics could build suitable initial solutions for CARP. The classical heuristics such as construct and strike, augment-merge, path scanning and Ulusoy's method are well-known methods for building initial CARP solutions (Corberán and Laporte 2013). In this study, the Ulusoy's method is applied to build the initial solution set. This method is a route-first, clusters-second heuristic (Ulusoy, 1985). First, it finds a giant tour covering all required edges in the graph ignoring capacity constraint and other similar constraints. Then, it partitions the giant tour into feasible sub-tours according to the ignored constraints in the first step. Unlike other classical heuristics, Ulusoy's method can build multiple distinct solutions on a specific graph via finding various giant tours (i.e., tours covering all required edges). The best solution within this set is chosen as the initial solution of the algorithm.

5.2. Finding a new solution

In each iteration of the algorithm, a new solution (t') is built via making some changes in the current solution (t). This is done by applying greedy removal and insertion heuristics. Removal heuristics omit the required edges which are not positioned in a suitable permutation of the edges, while the insertion heuristics seek to put the removed edges in better permutations. Let a, b, and c be three required edges that have been served sequentially in one trip by the same vehicle. If b is located in the shortest path of the two edges a and c, this edge has been placed in a good permutation of the edges. Then, it is not sensible to remove it from the permutation. Now, if the vehicle still has enough capacity to serve edge d, the best option is to put this edge in a permutation that imposes the minimum possible cost. Thereby, we first examine the possibility of placing d between two required edges that have been served in the tour. If the edge d could be located within the shortest path between the two edges, no extra cost is imposed. However, if this edge could not be placed between two required edges, it would be inserted in the permutation such that it imposes the minimum cost (Laporte et al., 2010).

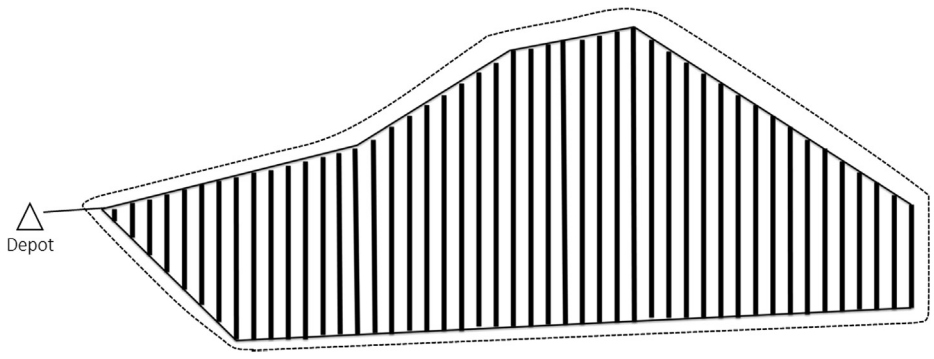


Fig. 3. Realistic field representation using ARP.

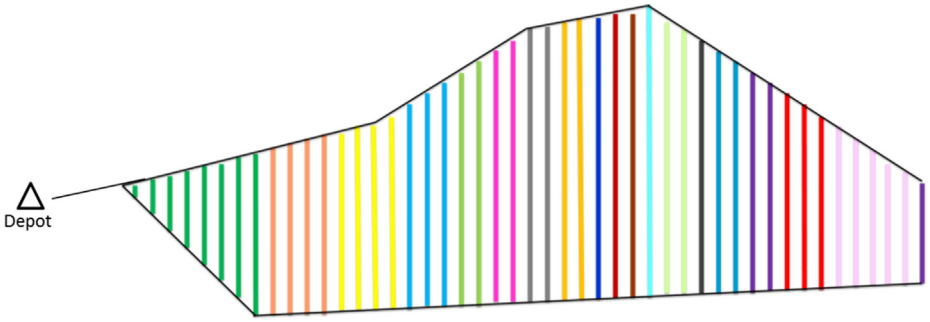


Fig. 4. Order of edge traversal using conventional harvesting pattern.

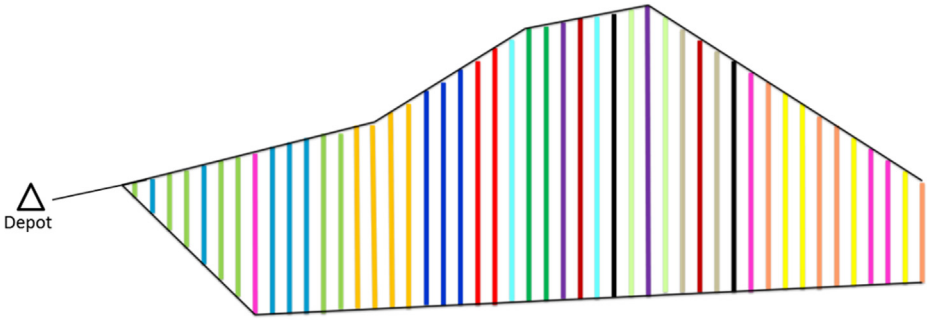


Fig. 5. Order of edge traversal using CARP approach and ALNS algorithm.

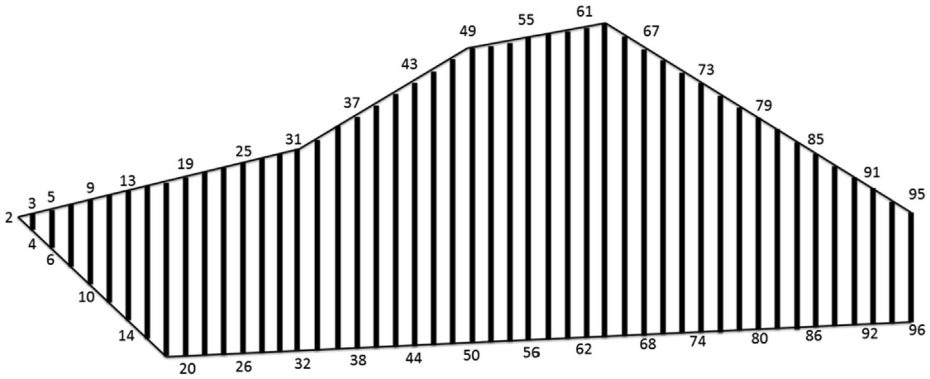


Fig. 6. The possible location of depot at each endpoint of the required edges.

5.3. Acceptance and stopping criteria

After finding the new solution (t') by applying the removal and insertion heuristics on the current solution (t), the new solution is evaluated using the objective function. If the new solution does not show any improvement in the value of the objective function, it is discarded. If the new solution does not show any improvement in the objective function, but its value is close to the value of the current solution $f(t')$, it included in the solutions set according to a predetermined amount of deviation (δ). Such solutions could be useful to avoid the local optimums. If the new solution has a better objective function value, it is accepted as the new best solution (t^*). This procedure is repeated until the stopping criterion is established. The stopping criterion is usually defined as a predetermined number of algorithm iterations, or algorithm exceeding its execution time limit. If the solution does not improve for a certain number of iterations, it will stop. The ALNS procedure is depicted in Fig. 2.

6. Computational results

The ALNS algorithm described above is coded in Matlab 2017a and run on Intel Corei5 CPU and 4 GB of RAMs to solve the in-filed or inter-field agricultural operations. Before solving the agriculture problem, the solution method is validated on a set of arc routing problem instances, including the well-known KSHS (Kiuchi et al, 1995) and GDB (Golden et al., 1983). Adaptive large neighborhood search can solve all the six instances of KSHS optimally (see Table 1). The GDB contains 23 instances. The proposed method obtains the optimal solution for six instances. The other instances are solved with a gap from the optimum. The observed gaps are between 2.48% and 8.97% (see Table 2) and on average, 4.46%. According to these results, the proposed method can obtain optimal or near-optimal results for practical problems.

Then, the order of path traversal for carrying out a harvesting operation in a realistic agricultural field is determined using the proposed method. In the realistic field, the conventional harvesting pattern using Ω-turns and the solution provided by the CARP method are compared. The field's graph consists of 96 nodes, 96 unrequired edges, and 47 required edges due to the implement width. Further, the field boundary and the position of the depot are shown in Fig. 3.

In a conventional harvesting pattern, the vehicle starts from the depot and serves the required edges one by one according to their order in the field, until the hopper capacity becomes full. After returning to

Table 3
Harvesting cost based on changing depot location.

Depot location	Harvesting Cost (Distance)	Depot location	Harvesting Cost (Distance)	Depot location	Harvesting Cost (Distance)
2	11,795	34	9261	66	8273
3	11,971	35	9249	67	8995
4	11,915	36	9062	68	8209
5	11,635	37	9624	69	9290
6	11,475	38	8844	70	8753
7	11,408	39	9158	71	9286
8	11,483	40	9000	72	8543
9	11,639	41	9457	73	9557
10	10,910	42	8655	74	8425
11	11,465	43	9191	75	9572
12	10,142	44	8705	76	8711
13	11,094	45	8631	77	10,058
14	10,555	46	8641	78	8827
15	10,529	47	9039	79	10,070
16	10,142	48	8207	80	8653
17	11,226	49	8956	81	10,363
18	10,350	50	8525	82	9053
19	10,972	51	8860	83	10,416
20	10,137	52	8110	84	9013
21	10,125	53	8718	85	10,591
22	10,094	54	8060	86	9649
23	10,068	55	8947	87	10,752
24	9899	56	7996	88	9697
25	10,456	57	8746	89	10,855
26	9848	58	8365	90	9726
27	10,047	59	8634	91	10,990
28	9644	60	8209	92	9787
29	9961	61	8767	93	11,212
30	9300	62	8215	94	9844
31	9330	63	8882	95	11,580
32	9288	64	8455	96	10,158
33	10,138	65	8813		

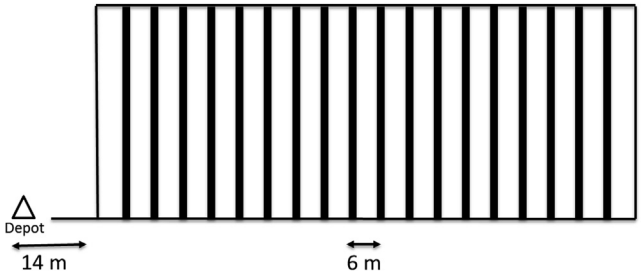


Fig. 8. Rectangular field graph, Working Width = 6 m.

the depot and unloading the hopper, the vehicle continues the job and harvests the remaining edges. Fig. 4 depicts the order of edge traversal using the conventional harvesting pattern in the field.

Due to its greedy nature, the conventional pattern is not necessarily the most efficient way to harvest the field. In fact, if the hopper capacity was small in comparison to the total harvest, this method would miss possible opportunities for improving the solution due to trying to traverse only the immediate required edge instead of other required edges which consume the remaining hopper capacity more efficiently. This would result in more trips back and forth between the field and the depot as the hopper becomes full faster. Hence, the CARP method is applied to the realistic field to find a better solution. Fig. 5 depicts the order of edge traversal using the CARP method. The solution obtained by the CARP method is 9.3% better than the conventional pattern result.

As suggested in previous studies, various factors can affect agricultural operations completion. During harvest operation, agricultural fleets meet the depot many times. Thus the location of the depot could directly affect the traversal cost. If the depot assumed as a temporary harvest facility, it could be located in different points of the headland. Based on this assumption, the effect of the depot location on the realistic field is discussed. The depot location is assumed at each endpoint

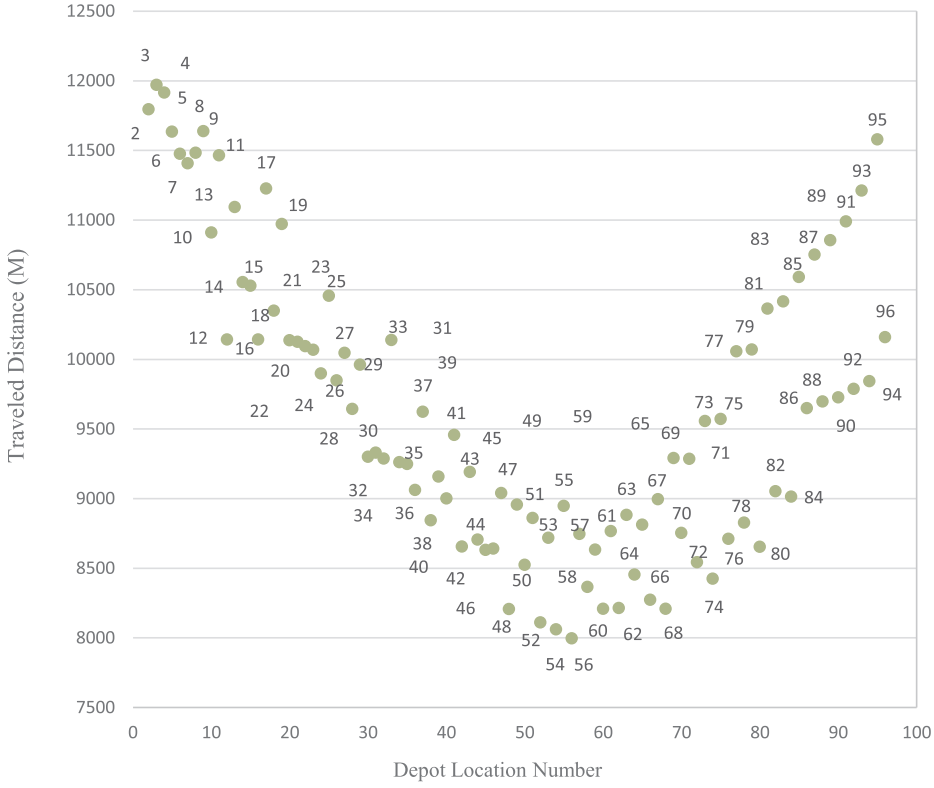


Fig. 7. Harvesting cost based on changing depot location.

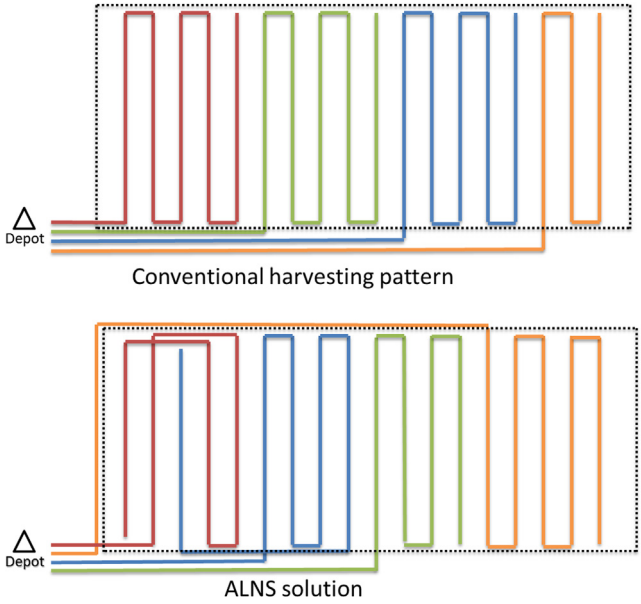


Fig. 9. Tours formed by conventional harvesting pattern and CARP method. Capacity = 1000 kg.

Table 4
Comparison of conventional harvesting pattern and ALNS result, Working Width = 6 m.

Capacity (Q)	# Tours	Conventional	ALNS	% Improvement
700	5	2304	2124	8.47
1000	4	2008	1876	7.04
1500	3	1776	1668	6.47
2500	2	1628	1592	2.26
3500	1	1324	1324	0.00

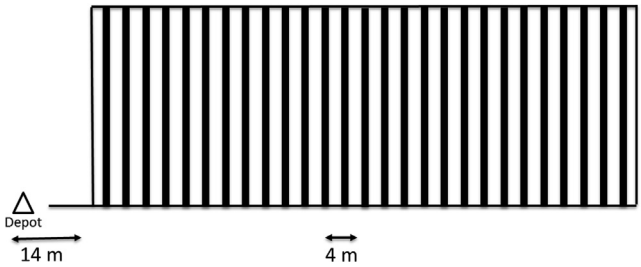


Fig. 10. A rectangular field graph, Working Width = 4 m.

Table 5
Comparison of conventional harvesting pattern and ALNS result, Working Width = 4 m.

Capacity (Q)	# Tours	Conventional	ALNS	% Improvement
700	5	2958	2774	6.63
1000	4	2498	2382	4.87
1500	3	2270	2222	2.16
2500	2	2118	2118	0.00
3500	1	1926	1926	0.00

of the required edges (Fig. 6). Then, the traversal order of them is determined using the proposed CARP method.

The depot location and associated distances provided by the ALNS algorithm is presented in Table 3. According to the results, the CARP method could provide more efficient routes than the conventional pattern. The results on this field show that as the depot location gets closer to the middle of the field, less harvesting cost is imposed (Fig. 7). Further, if the depot was located at node 56, the agricultural fleet would traverse the least total distance. The optimal routes and depot location could be planned by using CARP method for infield and interfiled agricultural operations.

7. Sensitivity analysis

The previous section presented the applicability of the CARP method for optimizing a realistic operation. Further, assuming the depot as a temporary facility, the notable role of the depot location in accomplishing an operation is confirmed. In the following, the effects of other factors such as vehicle capacity and field shape are investigated.

7.1. Effect of vehicle capacity

In order to examine the vehicle capacity effect, a rectangular field consisting of 18 required edges, due to the working width of 6 m, is assumed. The location of the depot and its distance to the field, as well as the working width of the vehicles, are depicted in Fig. 8.

In this examination, various capacities are considered for the vehicle, each resulting in a different number of tours. The order of edge traversal and tours formed by the conventional harvesting pattern and the CARP method, given that the capacity of the machine is 1000 kg, are shown in Fig. 9.

The results for different vehicle capacity are presented in Table 4. The number of tours needed to complete the operation, the total distance traveled by the conventional method, the total distances traveled by the CARP method, and the improvement percentages are also reported.

Moreover, the same examination is conducted, assuming the working width of 4 m. This assumption changes the filed graph. Now, the graph consists of 27 required edges (see Fig. 10). The corresponding results are presented in Table 5.

The ALNS algorithm could also provide more efficient solutions than the conventional pattern in the rectangular fields. Noticeably, when the vehicle's capacity exceeds the total demands, all the required edges of the field can be served just by a single tour. Therefore, in this case, there would not be a better pattern than the conventional pattern that the algorithm finds. Moreover, it is observed that generally, using vehicles with limited capacity (hence, requiring more tours) results in higher percentages of improvement when ALNS is used. Therefore, as more tours are needed to complete the operation, the ALNS algorithm could be more rewarding. Such an evaluation is also applicable to agricultural fleet selection before performing an operation.

7.2. Effect of field shape

Another factor, which could have considerable effects is the field shape. To examine the field shape effect, a rectangular-shaped field, with 18 identical edges (in terms of edge cost), is considered. Then the edges' costs and demands are changed (except in field A) drastically, to reach more complex-shaped fields. The edges costs and demands of field A are the same as the main rectangular field, but the position just shifted up or down on the field (see Fig. 11).

The order of edge traversal for each field is obtained by both methods. Additionally, the improvements provided using ALNS for the main field as wells as fields A–E are presented in Tables 4 and 6–10, respectively.

In almost all of the instances, the algorithm performs better than the conventional method. The average improvement for the main field is 6.06 and for fields A, B, C, D, and E it is equal to 7.39, 7.79, 11.18, 2.96, and 6.02, respectively. According to the obtained results, as a field has more different edges in terms of cost and demand (i.e., irregular shapes), there would be more potential for optimization using CARP. For the case of regular shapes including a circle and a hexagon, the amount of improvement is noticeably smaller. Thus, using CARP

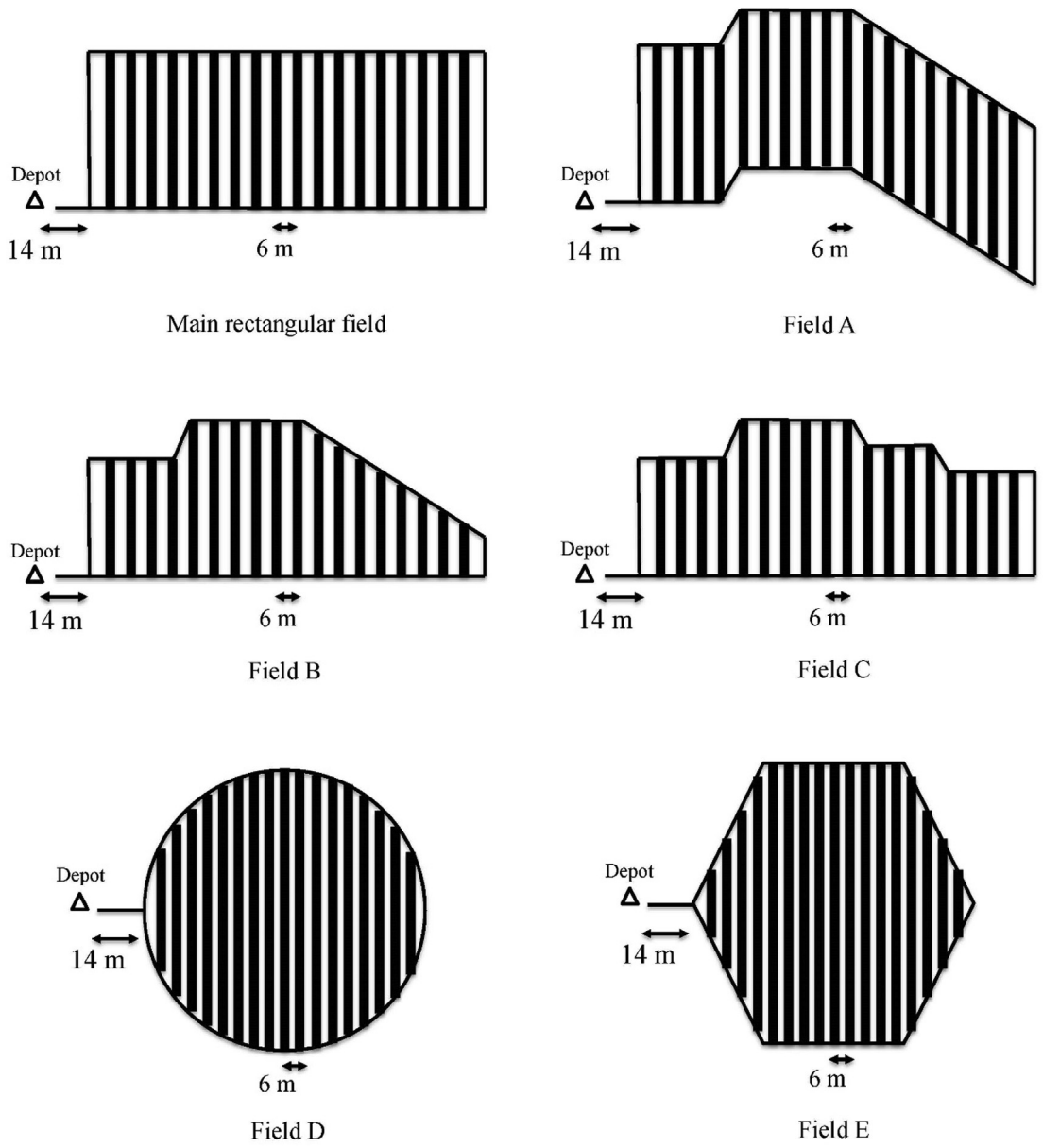


Fig. 11. Rectangular and more complex-shaped fields.

Table 6
Comparison of conventional harvesting pattern and ALNS result for field A.

Capacity (Q)	# Tours	Conventional	ALNS	% Improvement
700	5	2416	2224	8.63
1000	4	2096	1948	7.60
1500	3	1844	1724	6.96
2500	2	1676	1576	6.35

Table 7
Comparison of conventional harvesting pattern and ALNS result for field B.

Capacity (Q)	# Tours	Conventional	ALNS	% Improvement
700	4	1706	1504	13.43
1000	3	1409	1344	4.84
1500	2	1285	1232	4.30
2500	2	1304	1201	8.58

Table 8
Comparison of conventional harvesting pattern and ALNS result for field C.

Capacity (Q)	# Tours	Conventional	ALNS	% Improvement
700	4	1878	1552	21.01
1000	3	1499	1392	7.69
1500	2	1385	1304	6.21
2500	2	1364	1242	9.82

Table 9
Comparison of conventional harvesting pattern and ALNS result for field D.

Capacity (Q)	# Tours	Conventional	ALNS	% Improvement
700	5	2679	2679	0.00
1000	4	2560	2513	1.87
1500	3	2368	2225	6.43
2500	2	2147	2134.7	0.58

Table 10
Comparison of conventional harvesting pattern and ALNS result for field E.

Capacity (Q)	# Tours	Conventional	ALNS	% Improvement
700	5	2527	2538	−0.43
1000	4	2416	2159	11.90
1500	3	2250	1998	12.61
2500	2	2011	1899	5.90

method would be more applicable in complex-shaped fields, which consist of varied edges rather than rectangular- shaped ones.

8. Conclusion

Due to the various numbers of variables in agricultural operations, using conventional harvesting patterns is not necessarily the most efficient way. Completing these operations involves both monetary costs and time. If the operations are conducted more efficiently, fewer costs will be imposed. Therefore, optimizing agricultural operations matters.

In this study, following the logistical nature of the agricultural operations, the capacitated arc routing problem is introduced to plan the infield and inter-field operations efficiently. CARP, a combinatorial optimization problem, determines the agricultural fleets' paths in a field and finds the best order of traversing them. Contrary to the vehicle routing problem, implementing CARP on a field is relatively easy as no invalid or extra edge is created. This is an advantage of the proposed method as it does not require additional steps to deal with these edges.

As the capacitated arc routing problem is an NP-Hard problem, using metaheuristics is a viable option. Thus, an adaptive large neighborhood search was applied to obtain high-quality solutions. To verify the proposed method, two sets of classical problem instances were solved by the ALNS algorithm. Then, to demonstrate the application of CARP in the real world, a realistic harvesting operation problem was solved using both conventional harvesting pattern and CARP method. The comparison of the obtained results indicated that the CARP method could provide more efficient patterns to conduct the harvesting operation. Also, if a temporary facility is used as the depot in such operations, determining the depot location will be an important issue that directly affects the operation efficiency.

The results confirmed the success of the proposed approach in obtaining more efficient patterns. The potential for improving the operation using optimization increases, when more vehicles are required to fulfill the operation. This is true for more complex field shapes as well. This approach could be applicable in operation planning for both human-driven and autonomous vehicles. For future studies, the CARP could be extended assuming other constraints or conditions in agricultural fleet planning. For example, all input data of the proposed model are assumed deterministic and computable conveniently. However, estimating crop yield could be difficult and uncertain especially for agricultural products growing underground such as potato and beets. Therefore, the model could be developed to a stochastic CARP by assuming stochastic demands due to the uncertainties affecting the crops yield rates.

CRediT authorship contribution statement

Amin Khajepour: Conceptualization, Writing - original draft. **Majid Sheikhmohammady:** Supervision, Writing - review & editing. **Ehsan Nikbakhsh:** Methodology, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Ali, O., Verlinden, B., Van Oudheusden, D., 2009. Infield logistics planning for crop-harvesting operations. Eng. Optim. 41 (2), 183–197.
Arfini, F., Donati, M., Salazzo, R., Veneziani, M., 2016. Positive Mathematical Programming. In: Shrestha, S., Barnes, A., Ahmadi, B.V. (Eds.), Farm Level Modelling: Techniques, Applications and Policy. CABI, Boston, MA.
Bochtis, D.D., Sørensen, C.G., 2009. The vehicle routing problem in field logistics part I. Biosyst. Eng. 104 (4), 447–457.
Corberán, Á., Laporte, G. (Eds.), 2013. Arc Routing: Problems, Methods, and Applications. Society for Industrial and Applied Mathematics, Philadelphia PA.
Golden, B.L., Wong, R.T., 1981. Capacitated arc routing problems. Networks 11 (3), 305–315.
Golden, B.L., DeArmon, J.S., Baker, E.K., 1983. Computational experiments with algorithms for a class of routing problems. Comput. Oper. Res. 10 (1), 47–59.
Herrera-Cáceres, C., Pérez-Galarce, F., Álvarez-Miranda, E., Candia-Véjar, A., 2017. Optimization of the harvest planning in the olive oil production: A case study in Chile. Comput. Electron. Agric. 141, 147–159.
Jian, R., Malangmeih, L., Raju, S.S., Srivastava, S.K., Immaneulraj, K., Kaur, A.P., 2016. Optimization techniques for crop planning: a review. Indian J. Agric. Sci. 88 (12), 1826–1835.
Kiuchi, M., Shinano, Y., Hirabayashi, R., Saruwatari, Y., 1995. An exact algorithm for the capacitated arc routing problem using parallel branch and bound method. In Japanese In: Abstracts of the 1995 Spring National Conference of the Operational Research Society of Japan, pp. 28–29.
Laporte, G., Musmanno, R., Vocaturo, F., 2010. An adaptive large neighbourhood search heuristic for the capacitated arc-routing problem with stochastic demands. Transp. Sci. 44 (1), 125–135.
Oksanen, T., Visala, A., 2009. Coverage path planning algorithms for agricultural field machines. J. Field Rob. 26 (8), 651–668.
Pitakaso, R., Sethanan, K., 2019. Adaptive large neighborhood search for scheduling sugarcane inbound logistics equipment and machinery under a sharing infield resource system. Comput. Electron. Agric. 158, 313–325.
Ropke, S., Pisinger, D., 2006. An adaptive large neighborhood search heuristic for the pickup and delivery problem with time windows. Transp. Sci. 40 (4), 455–472.
Santoro, E., Soler, E.M., Cherri, A.C., 2017. Route optimization in mechanized sugarcane harvesting. Comput. Electron. Agric. 141, 140–146.
Sethanan, K., Neungmatcha, W., 2016. Multi-objective particle swarm optimization for mechanical harvester route planning of sugarcane field operations. Eur. J. Oper. Res. 252 (3), 969–984.
Seyyedhasani, H., Dvorak, J.S., 2017. Using the Vehicle Routing Problem to reduce field completion times with multiple machines. Comput. Electron. Agric. 134, 142–150.
Seyyedhasani, H., Dvorak, J.S., Roemmele, E., 2019. Routing algorithm selection for field coverage planning based on field shape and fleet size. Comput. Electron. Agric. 156, 523–529.
Srivastava, A.K., Goering, C.E., Rohrbach, R.P., Buckmaster, D.R., 1993. Engineering Principles of Agricultural Machines. American Society of Agricultural Engineers, St. Joseph, Michigan.
Spekken, M., de Bruin, S., 2013. Optimized routing on agricultural fields by minimizing maneuvering and servicing time. Precis. Agric. 14 (2), 224–244.
Ulusoy, G., 1985. The fleet size and mix problem for capacitated arc routing. Eur. J. Oper. Res. 22 (3), 329–337.
Zhou, K., Jensen, A.L., Bochtis, D.D., Sørensen, C.G., 2015. Simulation model for the sequential in-field machinery operations in a potato production system. Comput. Electron. Agric. 116, 173–186.
Zhou, K., Jensen, A.L., Sørensen, C.G., Busato, P., Bothtis, D.D., 2014. Agricultural operations planning in fields with multiple obstacle areas. Comput. Electron. Agric. 109, 12–22.